BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Functions and Return Values

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is a record?

2. Why retest a program after a fix?
   A) Check for new faults
   B) Prove no faults can exist
   C) Avoid testing old features
   D) Remove all input rules

3. "The Internet is a single computer containing every website." Correct this. Explain why.

4. A browser asks for a school's web page. "A client must host a website before asking for it." Correct this. Which computer provides the page?

Today's objectives

Define functions with parameters and return values. Use the technique in a different problem.

Key vocabulary - Read, cover, recall

Function
Named reusable subprogram that returns a value to its caller.

Parameters
Named inputs that allow a function to work on supplied arguments.

Return
A RETURN statement sends a calculated value to the calling code.

Modularity
Dividing a program into functions makes sections easier to reuse and test.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Function
Named reusable subprogram that returns a value to its caller. In the rectangle-area calculator, identify where this technique applies.

Parameters
Named inputs that allow a function to work on supplied arguments. In the taxi fare function, identify where this technique applies.

Return
A RETURN statement sends a calculated value to the calling code. In the test-mark average, identify where this technique applies.

Modularity
Dividing a program into functions makes sections easier to reuse and test. In the triangle area, identify where this technique applies.

I Do - Worked model

A rectangle-area calculator uses function and parameters. Explain what each contributes. [4 marks]

- Identify the program's purpose and the pieces of data needed.
- Function [1]; Named reusable subprogram that returns a value to its caller. [1].
- Parameters [1]; Named inputs that allow a function to work on supplied arguments. [1].
- Each feature needs a correct technical fact plus an explanation.

We Do - Guided practice

1. A taxi fare function needs function. Explain the purpose. [2 marks]

2. Explain parameters for a test-mark average. [2 marks]

3. When programming a triangle area, explain why return and modularity matter. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the return statement so the function calculates a pixel count.

```
function pixels(width, height)
return ...
endfunction
print(pixels(4, 3))
```

You Do - Independent exam questions

Q2. State TWO relevant ideas or operations from this topic. [2 marks]

Q3. Explain parameters and return in a shopping cost function. [4 marks]

Q4. A programmer makes a class total calculator. Explain the roles of function, return and modularity. [6 marks]

Exit Ticket - True or False?

A function is a named reusable subprogram that returns a value to its caller.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Named fields about one item or person. Fields may have different data types.

2. A - Check for new faults

3. A worldwide collection of interconnected computer networks.

4. A web server stores website resources and sends them when clients request them. Access needs working servers and network services.

I Do - Explain the model

Identify the program's purpose and the pieces of data needed.

Function [1]; Named reusable subprogram that returns a value to its caller. [1].

Parameters [1]; Named inputs that allow a function to work on supplied arguments. [1].

Each feature needs a correct technical fact plus an explanation.

We Do - Model answers

1. Function [1]; Named reusable subprogram that returns a value to its caller. [1].

2. Parameters [1]; Named inputs that allow a function to work on supplied arguments. [1].

3. Return [1]; A RETURN statement sends a calculated value to the calling code. [1]. Modularity [1]; Dividing a program into functions makes sections easier to reuse and test. [1].

Code First - Answer

```
function pixels(width, height)
return width * height
endfunction
print(pixels(4, 3))
```

Expected output: 12

You Do - Self-mark

Q2: Function [1]; Parameters [1].

Q3: Parameters [1]; Named inputs that allow a function to work on supplied arguments. [1]. Return [1]; A RETURN statement sends a calculated value to the calling code. [1].

Q4: Function [1]; Named reusable subprogram that returns a value to its caller. [1]. Return [1]; A RETURN statement sends a calculated value to the calling code. [1]. Modularity [1]; Dividing a program into functions makes sections easier to reuse and test. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.